

KELSO-LONGVIEW, WA, USAWMO: **727924**Lat: **46.117N**Lon: **122.894W**Elev: **20**StdP: **14.69**Time Zone: **-8.00 (NAP)**Period: **94-19**WBAN: **24223****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	23.4	27.0	16.0	12.3	29.5	20.6	15.5	30.6	21.8	46.1	19.5	46.6	1.2	140	0.509

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.8	88.3	67.3	83.7	66.2	80.6	64.7	69.0	84.6	67.2	81.8	65.4	78.4	8.8	350

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.1	86.4	75.4	61.3	81.0	72.6	60.2	77.9	71.1	32.9	84.8	31.6	81.3	30.1	78.6	78.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.9	14.7	12.6	DB	17.8	96.0	4.9	4.1	14.3	98.9	11.5	101.3	8.8	103.6	5.2	106.6	(4)
(5)			WB	16.5	72.8	4.7	2.6	13.1	74.7	10.4	76.2	7.8	77.7	4.4	79.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	52.6	41.2	42.5	46.5	50.3	56.3	60.8	66.1	66.4	61.8	52.8	45.6	40.3
	DBStd	10.61	5.78	5.38	4.95	5.15	5.53	5.35	4.90	4.55	5.00	5.03	5.77	5.99
	HDD50	1167	277	213	131	54	8	0	0	0	0	28	152	304
	HDD65	4732	738	630	573	440	275	149	42	37	122	379	581	766
	CDD50	2124	3	3	24	65	204	324	500	508	353	115	22	3
	CDD65	213	0	0	0	1	6	23	78	79	25	1	0	0
	CDH74	2568	0	0	2	42	159	334	831	810	377	13	0	0
	CDH80	863	0	0	0	10	42	111	296	291	112	1	0	0
Wind	WSAvg	5.0	5.7	4.8	5.2	4.7	4.7	4.9	5.3	4.9	4.2	4.2	5.4	5.8
Precipitation	PrecAvg	46.3	6.6	4.8	4.5	3.7	2.8	2.0	0.9	1.5	2.2	4.0	7.0	7.1
	PrecMax	57.7	13.6	11.1	8.5	7.2	6.2	4.8	3.2	5.9	5.0	7.9	14.8	14.3
	PrecMin	34.5	0.3	0.2	0.6	1.2	0.4	0.5	0.0	0.0	0.1	0.3	1.1	2.2
	PrecStd	6.7	3.3	2.5	1.8	1.5	1.4	1.0	0.8	1.4	1.4	2.0	3.0	2.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	57.1	61.1	71.7	80.6	85.1	90.6	94.0	93.3	89.9	74.9	63.1	55.4
		MCWB	52.3	49.1	54.4	59.9	63.1	68.3	70.6	68.4	65.6	59.9	57.4	52.2
	2%	DB	54.1	56.7	63.9	71.9	78.9	82.1	87.9	87.8	82.3	69.7	58.9	53.7
		MCWB	50.5	48.5	52.1	56.6	61.6	64.8	68.2	67.4	64.7	58.6	53.9	50.9
	5%	DB	51.8	54.1	59.2	65.3	72.5	77.0	82.1	82.0	78.8	64.4	55.5	50.4
		MCWB	48.7	47.9	49.5	53.2	59.0	62.4	66.4	66.2	62.8	56.5	51.9	47.7
	10%	DB	48.5	51.7	55.1	60.9	67.6	72.3	78.8	78.6	73.0	61.4	54.2	48.0
		MCWB	45.7	47.1	48.6	50.9	56.9	60.3	64.5	64.6	61.0	55.4	51.0	45.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	54.5	53.1	56.7	61.3	66.1	69.6	72.8	71.1	68.5	63.3	58.8	53.6
		MCDB	56.1	56.9	66.6	76.5	82.0	87.3	90.7	88.4	84.0	69.4	62.1	55.0
	2%	WB	51.1	50.7	53.3	57.1	62.4	65.9	69.3	68.7	66.0	60.1	55.3	50.6
		MCDB	52.8	54.4	60.1	68.4	75.3	80.9	84.7	83.8	80.0	66.1	58.2	52.6
	5%	WB	48.3	48.7	51.4	54.2	60.0	63.1	67.1	66.9	63.9	57.9	53.2	48.0
		MCDB	50.6	52.6	57.6	62.8	70.7	75.7	81.2	81.2	75.7	63.1	55.2	49.7
	10%	WB	46.3	46.6	49.7	52.1	57.4	60.8	65.0	65.0	61.9	56.1	51.3	46.1
		MCDB	48.1	49.7	55.3	58.6	66.2	70.7	77.3	77.1	71.5	60.8	53.7	47.4
Mean Daily Temperature Range	5% DB	MDBR	10.2	14.3	16.0	18.0	19.0	19.0	21.8	21.9	21.7	16.5	11.7	9.5
		MCDBR	12.9	20.5	24.5	29.9	30.4	29.9	30.6	30.8	32.0	23.5	14.2	10.7
		MCWBR	9.1	13.2	14.1	16.0	16.0	15.2	14.8	14.5	15.9	13.8	10.0	8.1
	5% WB	MCDBR	10.3	15.0	19.9	24.9	26.4	26.5	27.9	27.4	27.6	18.2	10.8	8.7
MCWBR		8.1	10.4	12.3	14.1	14.4	14.2	14.2	13.6	14.2	11.7	8.3	7.3	
Clear-Sky Solar Irradiance	taub	0.324	0.316	0.333	0.363	0.378	0.367	0.353	0.358	0.344	0.334	0.324	0.329	
	taud	2.504	2.548	2.509	2.390	2.370	2.419	2.478	2.472	2.510	2.539	2.523	2.435	
	Ebn at Noon	247	276	285	284	282	285	288	282	278	264	245	228	
	Edh at Noon	21	24	29	36	37	36	33	32	29	24	21	20	
All-Sky Solar Radiation	RadAvg	350	616	899	1314	1669	1838	2072	1772	1323	763	411	289	
	RadStd	57	87	118	123	154	178	151	128	119	93	55	43	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (38 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+45

Nomenclature: See separate page